

## SEMESTER S8

### DIGITAL IMAGE PROCESSING

|                                            |                                    |                    |                |
|--------------------------------------------|------------------------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>PEEVT 864</b>                   | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0                            | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3                                  | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | Signal Processing<br>(Course code) | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To study the image fundamentals and transforms necessary for image processing.
2. To familiarize different image processing techniques

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                  | <b>Contact Hours</b> |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Digital Image Fundamentals-Elements of visual perception, image sensing and acquisition, image sampling and quantization, basic relationships between pixels – neighborhood, adjacency, connectivity, distance measures. Brightness, contrast, hue, saturation, mach band effect, Colour image fundamentals-RGB, CMY, HIS models, 2D sampling, quantization. | <b>9</b>             |
| <b>2</b>          | Image Enhancement: Spatial domain methods: point processing-intensity transformations, histogram processing, image subtraction, image averaging, geometric transformation Sharpening filters – first and second derivative, two-dimensional DFT and its inverse, frequency domain filters – low-pass and high-pass.                                          | <b>9</b>             |
| <b>3</b>          | Image segmentation: Classification of Image segmentation techniques, region approach, clustering techniques. Classification of edges, edge detection Image restoration: Restoration Models, Linear Filtering Techniques: Inverse and Wiener                                                                                                                  | <b>9</b>             |
| <b>4</b>          | Image Compression- Need for compression, redundancy, classification of image compression schemes,–inter-pixel and psycho-visual; Lossless compression – predictive, Lossy compression- predictive and transform coding –DCT, (basics only); Still image compression standards – JPEG .                                                                       | <b>9</b>             |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| Attendance | Assignment/<br>Microproject | Internal<br>Examination-1<br>(Written) | Internal<br>Examination- 2<br>(Written ) | Total |
|------------|-----------------------------|----------------------------------------|------------------------------------------|-------|
| 5          | 15                          | 10                                     | 10                                       | 40    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| Part A                                                                                                                                                                                            | Part B                                                                                                                                                                                                                                                                                                             | Total     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b> |

**Course Outcomes (COs)**

At the end of the course students should be able to:

| Course Outcome |                                                                                 | Bloom's<br>Knowledge<br>Level (KL) |
|----------------|---------------------------------------------------------------------------------|------------------------------------|
| <b>CO1</b>     | Understand the various concepts and mathematical transforms for image transform | <b>K2</b>                          |
| <b>CO2</b>     | Interpret the various image enhancement techniques                              | <b>K2</b>                          |
| <b>CO3</b>     | Illustrate the image segmentation algorithm                                     | <b>K2</b>                          |
| <b>CO4</b>     | Summarise basic image compression techniques                                    | <b>K2</b>                          |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

**CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)**

|            | PO1      | PO2      | PO3 | PO4 | PO5      | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12     |
|------------|----------|----------|-----|-----|----------|-----|-----|-----|-----|------|------|----------|
| <b>CO1</b> | <b>3</b> | <b>1</b> |     |     |          |     |     |     |     |      |      | <b>2</b> |
| <b>CO2</b> | <b>3</b> | <b>2</b> |     |     | <b>2</b> |     |     |     |     |      |      | <b>2</b> |
| <b>CO3</b> | <b>3</b> | <b>2</b> |     |     | <b>2</b> |     |     |     |     |      |      | <b>2</b> |
| <b>CO4</b> | <b>3</b> | <b>2</b> |     |     | <b>2</b> |     |     |     |     |      |      | <b>2</b> |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                          |                                                |                              |                         |
|-------------------|--------------------------|------------------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b> | <b>Name of the Author/s</b>                    | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | Digital Image Processing | Gonzalez Rafel C                               | Pearson Education            | 2009                    |
| 2                 | Digital Image Processing | S Jayaraman,<br>S Esakkirajan,<br>T Veerakumar | Tata Mc Graw Hill            | 2015                    |

| <b>Reference Books</b> |                                          |                             |                              |                         |
|------------------------|------------------------------------------|-----------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                 | <b>Name of the Author/s</b> | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                      | Fundamentals of Digital Image Processing | Jain Anil K                 | PHI                          | 1988                    |
| 2                      | Digital image processing                 | Kenneth R Castleman         | Pearson Education,           | 2/e,2003                |
| 3                      | Digital Image Processing                 | Pratt William K             | John Wiley                   | 4/e,2007                |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                           |
|---------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Module No.</b>                     | <b>Link ID</b>                                                                            |
| <b>1</b>                              | <a href="https://nptel.ac.in/courses/117105135">https://nptel.ac.in/courses/117105135</a> |
| <b>2</b>                              | <a href="https://nptel.ac.in/courses/117105135">https://nptel.ac.in/courses/117105135</a> |
| <b>3</b>                              | <a href="https://nptel.ac.in/courses/117105135">https://nptel.ac.in/courses/117105135</a> |
| <b>4</b>                              | <a href="https://nptel.ac.in/courses/117105135">https://nptel.ac.in/courses/117105135</a> |